Theorem1.6 (Right Cancellation Law)

Let $A, B,$ and C be square matrices of order n. If A is non-singular and $BA = CA$, then $B = C$.

Proof

Since A is non-singular, A^{-1} exists and $AA^{-1} = A^{-1}A = I_n$. Taking $BA = CA$ and post-multiplying both sides by A^{-1}, we get $(BA)A^{-1} = (CA)A^{-1}$. By using the associative property of matrix multiplication and property of inverse matrix, we get $B = C$. ■

Note

If A is singular and $AB = AC$ or $BA = CA$, then B and C need not be equal. For instance, consider the following matrices:

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \text{ and } C = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}.$$

We note that $|A| = 0$ and $AB = AC$; but $B \neq C$.

Theorem 1.7 (Reversal Law for Inverses)

If A and B are non-singular matrices of the same order, then the product AB is also non-singular and $(AB)^{-1} = B^{-1}A^{-1}$.

Proof

Assume that A and B are non-singular matrices of same order n. Then, $|A| \neq 0$, $|B| \neq 0$, both A^{-1} and B^{-1} exist and they are of order n. The products AB and $B^{-1}A^{-1}$ can be found and they are also of order n. Using the product rule for determinants, we get $|AB| = |A||B| \neq 0$. So, AB is non-singular and

$$(AB)(B^{-1}A^{-1}) = (A(BB^{-1}))A^{-1} = (AI_n)A^{-1} = AA^{-1} = I_n;$$
$$(B^{-1}A^{-1})(AB) = (B^{-1}(A^{-1}A))B = (B^{-1}I_n)B = B^{-1}B = I_n.$$

Hence $(AB)^{-1} = B^{-1}A^{-1}$. ■

Theorem 1.8 (Law of Double Inverse)

If A is non-singular, then A^{-1} is also non-singular and $(A^{-1})^{-1} = A$.

Proof

Assume that A is non-singular. Then $|A| \neq 0$, and A^{-1} exists.

Now $|A^{-1}| = \dfrac{1}{|A|} \neq 0 \Rightarrow A^{-1}$ is also non-singular, and $AA^{-1} = A^{-1}A = I$.

Now, $AA^{-1} = I \Rightarrow \left(AA^{-1}\right)^{-1} = I \Rightarrow \left(A^{-1}\right)^{-1} A^{-1} = I$. ... (1)

Post-multiplying by A on both sides of equation (1), we get $\left(A^{-1}\right)^{-1} = A$. ■

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